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UrbanPulse AERVIO RA

Turns a routine weather check into personalised risk guidance

IEEE CURTIN ASPIRE 2026 • TEAM D



Project content by the team; AI-assisted design

IEEE CURTIN
ASPIRE 26
— INNOVATE | IMPACT | INSPIRE —

A personal impact.



Jean Sybico & Nena Omalde

Nathaniel's mother & grandmother



The struggle

- Extreme heat affects them every day
- Asthma makes it even harder
- They know their limitations
- Still has to check so many apps
- Still has to manually calculate the decision



It's not just them

- Millions face the same struggle
- People with asthma, allergies, or other medical conditions
- **Time and energy is wasted** searching for information and making a decision

THE PROBLEM

Data exists. Clarity doesn't.



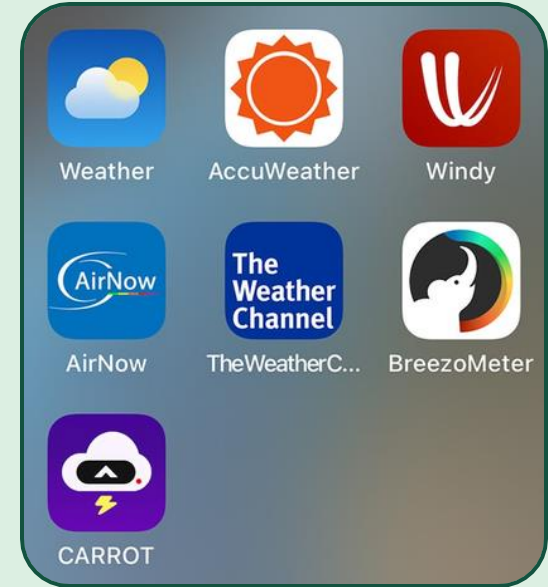
HEAT IS DANGEROUS

- 489,000 people die from heat every year [1].
- In the UAE, 86% of adults have felt heat exhaustion, 43% heatstroke [2].



MANY FACTORS

- Heat is only one factor
- Humidity, dust, pollen, and air pollution add to the risk [3].



SPREAD ACROSS APPS

- The data isn't missing
- It's **scattered across separate** apps as raw, impersonal numbers

Less autonomy. More risk. More guesswork.



LOSS OF AUTONOMY

- They can decide for themselves, but it takes real **repeated effort**
- Instead, many **stay in or rely on others**



HIGHER HEALTH RISK

- Misjudging heat exposure **causes real harm**
- Flare-ups, exhaustion, or worse [4].



DECISION BURDEN

- They check two or three apps
- And still have to **work it out themselves**
- Seniors have a difficult time checking so many apps

So many apps. You're still the one connecting the dots.



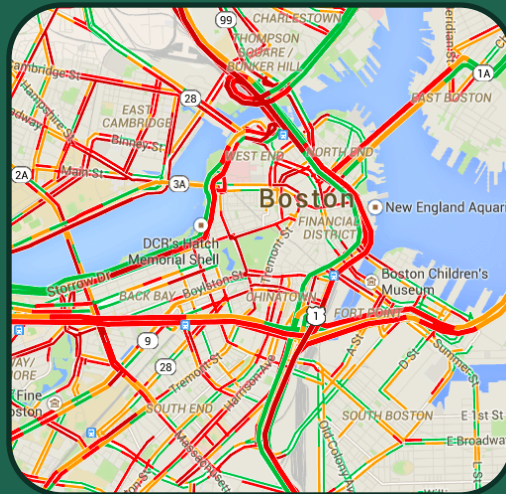
Air Quality Apps

Only reports local air quality



Weather Apps

Shows weather conditions



We solved this for traffic already:

- Google Maps puts the **when**, **where**, and **how bad** the traffic is on one view so we can plan around it
- **Nobody does that for the conditions outside**



AirRater [5]

Tracks personal symptom patterns



ClimApp [6]

Personalised thermal stress guidance

OUR SOLUTION



HOW IT WORKS

Transforming data to a decision you can act on.

What Aerviora Does:

1. Tell us your context
Activity · Duration · Sensitivities

2. We read the environment
Weather · Heat · Air quality · UV

3. Get guidance you can act on
Concern · Action · Reasoning · Timing

Plan around conditions, rather than react to them.

DEMO OF PROTOTYPE

Aerviora

PROTOTYPE

Know your lower-risk time outside.

Aerviora brings heat, air quality, dust and pollen into one clear, explainable outdoor decision.

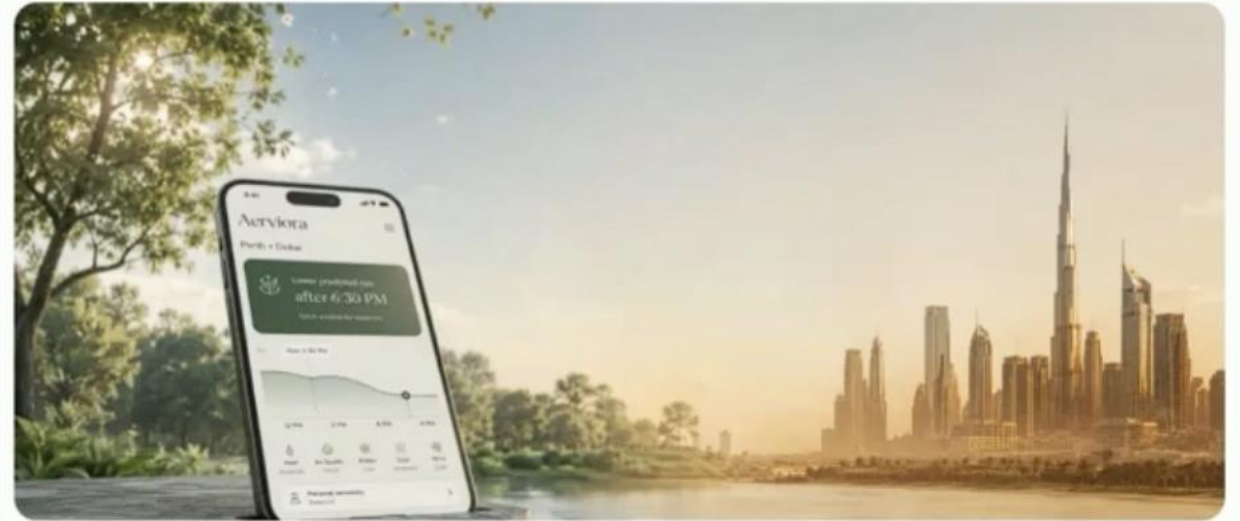
Start an outdoor check

Privacy Dashboard

No detailed medical history required.

• CONCEPT VISUAL

Perth to Dubai — one adaptable decision framework



One clear decision from complex environmental signals

1

2

3

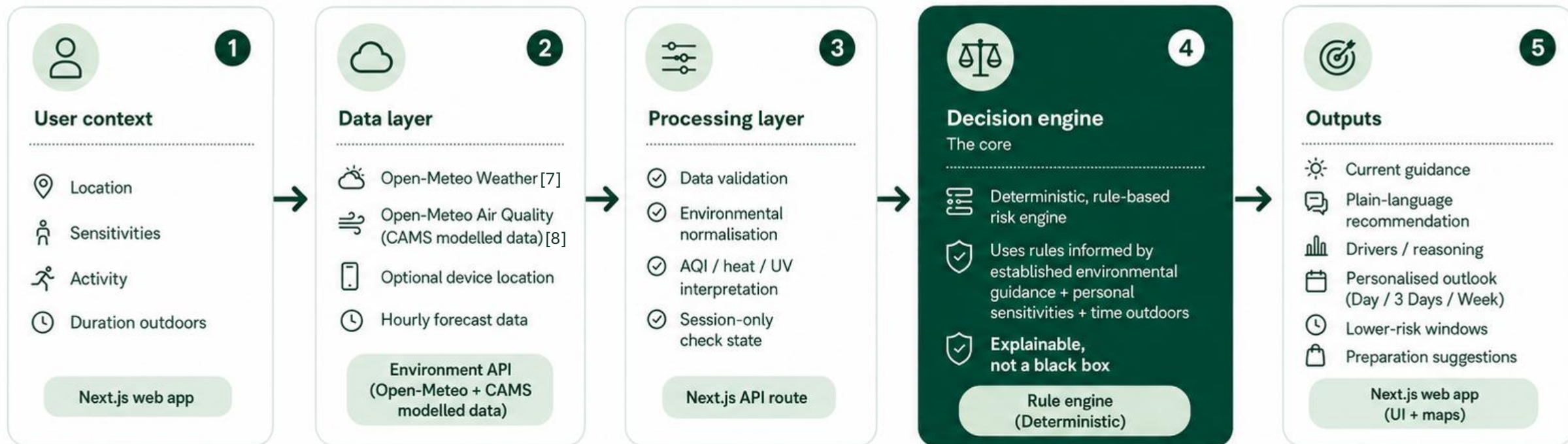
From environmental data to personalised guidance.

Aerviora turns trusted public data and your context into clear, actionable guidance.



Built on trusted sources.

Deterministic. Transparent.
Designed to support outdoor decisions.



Privacy-first

We do not store health profiles.
You stay in control.



No account required

Prototype works out of the box.



Session-only data

Context used only for this session.



Optional local map

See conditions around you.



Demo scenarios

Preloaded cases for presentations.



Why rule-based?

- ✓ Transparent reasoning
- ✓ Health-affecting guidance should be explainable



Built with Next.js, TypeScript and open environmental data APIs.

Public data. Deterministic logic. Clear answers.

WHAT MAKES US UNIQUE

What makes Aerviora unique?

FEATURE	WEATHER	AQI / POLLEN	ALERTS	AERVIORA
Multiple environmental inputs	✓	Partial	Partial	✓
User's sensitivities profile	—	—	—	✓
Activity-aware	—	—	—	✓
Suggests a lower-risk time	—	—	—	✓
Explains risk	—	Limited	Limited	✓
Learns from usage over time	—	—	—	✓

Designed for everyday decisions not only emergency alerts.

What is our innovation?

1

Consolidation:

One view across every condition, instead of one app for each.

2

Timing:

Most apps tell you there is a heatwave today, Aerviora tells you which hours are worse and what to do about them.

3

Personalisation:

No generic numbers. Aerviora adjusts its guidance to fit the person reading it.

What is our impact on the SDGs?



- People can **avoid health risks** by making information easily accessible
- Helps people avoid **underestimating danger** and getting hurt
- Stops people **overestimating risk** and missing out on days that were actually fine



- A liveable city is one people can move through with confidence
- **Gives confidence** to move through the city
- Help **residents to adapt** their daily activities to environmental conditions of the city



- The infrastructure already exists:
 - City monitoring networks
 - Government published extreme-weather plans [9]
- **Aerviora can put all these systems and data in one place**

COST STRUCTURE

Cost structure & revenue streams

Sustainable unit economics. Scalable impact.



COST STRUCTURE



Environmental data APIs [10]

(weather, air quality, dust, pollen)

FREE

for prototyping

\$30–\$100

/ month
as usage scales



Backend hosting [11]

(fetch, cache, send data)

\$50–\$300

/ month
at early-stage traffic



Ongoing engineering

development, maintenance,
testing beyond initial build



Medical expert review

consulting basis to review
risk-assessment thresholds per profile



Privacy and security compliance

data protection, audits, compliance
and ongoing monitoring



REVENUE STREAMS



Freemium (B2C) [12]

free tier with basic conditions + ads;
paid tier adds deeper personalisation

\$2–\$5 / month
or **\$20–\$30** / year



Employer access [13]

staff benefit for outdoor workers

\$2–\$10

/ worker / month



Institutional access [14]

universities, councils,
health organisations

\$0.75–\$3

/ person / year



Business offers / discounts [15]

Aerviora takes a
cut per sale

5–20%

of sale value



Public-health grants / sponsorships [16]

comparable health apps average approx.

\$345,000

in grants



Future — anonymised data licensing

aggregated environmental-health
trend data, not personal data

Long-term
high-margin opportunity

FUTURE PLANS

Our goals in the future.

PHASE 1

- Getting **more sources** for data
- Improve data accuracy by combining different sources

PHASE 2

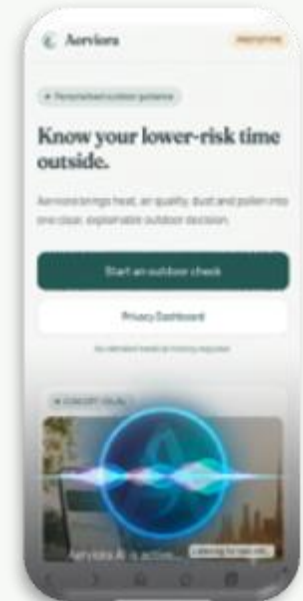
- Partnerships with **medical professionals**
- Improve the recommendation engine
- Get professional advice

PHASE 3

- A pilot with:
 - **employers of outdoor workforces**
 - **aged care providers**
 - **students**
- Put Aerviora in the hands of the people who need it most

PHASE 4

- Let Siri, Alexa, and other **assistants** pull from Aerviora



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Thank You!



MEET THE TEAM



<https://aerviora-private.vercel.app/>



Daehwan Yeo

Computing (Honours)
Curtin Bentley



Erdzean Nathaniel Sybico

Mechanical Engineering (Honours)
Curtin Dubai



Aamina Rozan

Cyber Security
Curtin Colombo



Shamas Mohamed Sulaiman

Mechatronics Engineering
Curtin Colombo



Chun Yiew Hii

Electrical and Electronic Engineering
Curtin Malaysia